

LEXINGTON, NE, USA

WMO: 725624

Lat: 40.789N

Lon: 99.771W

Elev: 2405

StdP: 13.46

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 14994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	3.5	-8.1	3.9	0.4	-3.2	5.1	5.7	32.7	37.0	28.7	34.8	7.1	300	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	95.1	71.8	91.4	71.4	89.3	71.2	76.5	87.5	74.9	86.0	73.3	84.7	13.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	134.2	83.1	71.8	128.6	82.1	70.0	120.7	80.2	41.7	87.9	40.2	85.4	38.6	85.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.3	25.6	22.3	-11.3	100.5	5.3	3.9	-15.1	103.3	-18.2	105.6	-21.2	107.8	-25.0	110.7		
			-11.9	79.7	5.1	1.9	-15.6	81.0	-18.7	82.2	-21.5	83.2	-25.3	84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	27.4	29.8	40.4	49.3	59.6	70.4	75.6	72.7	65.1	51.1	39.5	28.3
	DBStd	19.30	10.96	11.46	10.93	9.34	8.46	6.87	5.51	5.66	8.34	9.24	9.35	10.21
	HDD50	2833	701	566	327	122	15	0	0	0	3	96	331	672
	HDD65	6080	1166	985	762	476	210	27	4	8	102	438	766	1136
	CDD50	3154	0	2	31	100	314	611	793	702	456	130	15	0
	CDD65	925	0	0	0	4	43	188	332	246	104	8	0	0
	CDH74	10157	0	1	38	134	632	2090	3549	2337	1209	158	9	0
	CDH80	4188	0	0	6	29	214	849	1621	958	474	37	0	0
(14) Wind	WSAvg	9.7	9.2	10.1	11.1	12.5	11.1	10.1	8.3	7.5	9.0	9.5	9.3	9.1
(15) Precipitation	PrecAvg	23.2	0.4	0.6	1.1	2.6	3.6	3.8	3.1	3.0	1.8	1.8	0.8	0.5
	PrecMax	34.2	1.4	1.5	2.8	7.0	10.1	8.1	9.4	5.6	4.3	5.9	3.3	3.1
	PrecMin	10.7	0.0	0.0	0.0	0.6	0.8	0.7	0.1	0.8	0.3	0.0	0.0	0.0
	PrecStd	4.7	0.3	0.4	0.8	1.5	2.1	1.9	2.1	1.3	1.0	1.4	0.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	68.4	78.8	83.7	91.1	97.1	99.6	98.6	95.0	84.4	74.7	61.1
		MCWB	45.4	49.5	55.1	58.4	64.0	69.9	71.6	71.3	68.1	61.2	53.5	45.2
	2%	DB	54.3	59.4	72.1	78.0	85.9	92.7	95.3	92.7	90.2	78.7	67.9	54.5
		MCWB	41.9	44.4	52.6	57.5	64.8	70.6	73.5	72.6	68.2	59.4	51.0	43.0
	5%	DB	48.1	53.2	65.5	72.5	81.2	89.2	91.5	89.3	85.9	73.1	61.5	48.2
		MCWB	39.0	42.3	50.2	55.7	62.8	70.5	73.2	72.7	67.1	57.0	47.8	39.3
	10%	DB	42.9	46.3	59.4	67.2	76.6	84.7	89.3	85.7	81.3	68.2	55.4	43.2
		MCWB	36.0	38.8	47.2	53.5	61.6	68.7	73.0	71.8	64.8	55.1	45.4	36.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	50.3	60.0	62.8	70.8	76.6	79.3	77.8	73.3	65.9	56.6	48.9
		MCDB	60.3	64.5	68.4	75.7	82.5	87.0	90.1	88.6	86.1	77.4	66.5	59.2
	2%	WB	42.1	46.0	55.4	59.5	67.7	73.8	77.0	75.9	71.1	62.3	52.6	42.9
		MCDB	53.0	57.6	66.9	72.9	79.0	86.3	88.4	86.5	84.1	71.9	63.6	53.8
	5%	WB	39.3	42.5	51.4	56.9	65.4	72.1	75.5	74.3	69.1	59.6	49.5	39.6
		MCDB	46.8	52.1	63.9	69.5	77.1	84.7	86.8	84.7	80.7	69.3	60.8	47.5
	10%	WB	36.3	39.1	48.0	54.4	63.1	70.2	74.0	72.6	67.2	56.8	45.9	36.5
		MCDB	42.2	46.7	59.7	65.7	73.7	82.4	85.6	82.7	78.0	66.9	56.2	42.7
(35) Mean Daily Temperature Range	5% DB	MDBR	22.5	22.4	25.3	24.6	23.7	23.3	22.9	22.5	26.2	25.9	25.8	22.2
		MCDBR	31.5	33.3	37.8	34.8	30.8	28.8	27.7	27.8	31.0	34.7	35.8	31.1
		MCWBR	20.2	20.3	20.6	18.0	14.0	12.0	11.9	11.7	12.9	17.4	20.6	20.3
	5% WB	MCDBR	29.5	31.1	33.4	30.3	25.5	24.8	23.6	23.8	25.8	27.3	33.6	29.2
		MCWBR	19.6	19.7	19.5	16.9	13.4	12.3	12.2	12.2	12.6	15.9	20.0	19.9
(40) Clear-Sky Solar Irradiance	taub		0.269	0.273	0.298	0.328	0.358	0.375	0.391	0.384	0.352	0.311	0.274	0.264
	taud		2.507	2.508	2.466	2.402	2.395	2.382	2.377	2.384	2.427	2.519	2.540	2.514
	Ebn at Noon		288	303	304	299	291	285	279	279	281	284	284	280
	Edh at Noon		23	27	31	36	37	38	38	36	33	26	22	21
(44) All-Sky Solar Radiation	RadAvg		720	989	1349	1681	1943	2234	2201	1910	1548	1107	795	621
	RadStd		62	71	111	93	128	134	86	97	89	111	53	50

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page